

RAGY NOUR

Unity Game Developer

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PROFESSIONAL SUMMARY

Performance-driven Unity Developer with 4+ years of experience shipping mobile games to Google Play. Specialized in GPU/CPU optimization (Instancing, Compute Shaders, MaterialPropertyBlock), modular game architecture using SOLID principles and design patterns, and custom editor tooling that accelerated level production by 90%. Shipped titles reaching 1,000+ organic users with stable 60 FPS on low-end Android devices. Background in Electronics Engineering with strong foundations in linear algebra, data structures, and signal processing.

TECHNICAL SKILLS

Engine	Unity (3D), URP, Editor Scripting, Addressables, Timeline, Cinemachine, New Input System
Languages	C# (advanced), HLSL/Cg (Shaders & Compute), Python, JSON/CSV parsing
Architecture	SOLID, OOP, Design Patterns (Observer, Command, State, Factory, Object Pooling, Singleton)
Optimization	GPU Instancing, Draw Call Batching, MaterialPropertyBlock, Lazy Loading, Unity Profiler, Frame Debugger, Memory Profiler
Graphics	Custom Shaders (Vertex/Fragment), Compute Shaders, Animation Texture Baking, Shader Graph, URP Customization, VFX/Particle Systems
AI & Gameplay	Finite State Machines, NavMesh, Physics-based AI, Procedural Animation (IK), Raycasting, Trajectory Prediction, Touch Input Systems
Animation	Animator/Mecanim, Inverse Kinematics, Animation Rigging, DOTween, Procedural Animation, Runtime Mesh Baking
Backend	Firebase (Auth, Remote Config, Crashlytics), REST APIs, AdMob/Unity Ads, Google Play Console
Tools	Git (GitHub), Blender (3D Modeling), Visual Studio, Figma, Android SDK, ASO
Platforms	Android (primary), Windows, WebGL

SELECTED PROJECTS

Swipe Dash - Action Mobile Game

Unity, C#, URP, Cinemachine, New Input System

Solo Developer · In Development · [Demo Video](#)

- Swipe-to-Dash Combat:** Designed a touch-based path-drawing system using screen-to-world raycasting and dot-product direction filtering - players draw attack paths that the character executes as chained dashes through enemies.
- Game Feel & Juice:** Implemented hit-stop (TimeScale manipulation), Cinemachine camera shake, ghost trail via runtime SkinnedMesh baking with fading materials, dash ground lines (LineRenderer + DOTween fade), and VFX particle hit feedback.
- Enemy AI:** Built FSM-based enemies with range-based state transitions (Physics.CheckSphere), smooth pursuit, attack cooldowns, and per-instance hit-flash via MaterialPropertyBlock. Extended via inheritance for ranged Archer variant with projectile attacks.
- Performance:** Used Object Pooling for particles and projectiles, New Input System for unified touch/gamepad support, and coroutine-driven dash sequencing for zero-allocation combat loops.

Football Strike - 3rd-Person Mobile Sports Game

Unity, C#, HLSL, Blender

Solo Developer · Released on Android · [Demo Video](#)

- GPU Crowd System:** Replaced SkinnedMeshRenderers with GPU Instancing + Animation Texture Baking, rendering 2,000+ crowd fans in <3 draw calls - reduced rendering overhead by ~85%.
- Physics-Based AI:** Programmed goalkeeper AI that calculates ball velocity trajectories to determine intercept positions, blended with FSM logic for positioning and man-marking behaviors.
- Procedural Animation (IK):** Implemented Inverse Kinematics to dynamically align player limbs to ball contact points during high-speed interactions, eliminating visual-physics discrepancies.
- Manager Mode:** Built a data-driven dashboard with serialized save states for team lineups, player transfers, economy system, and unlockable content progression.
- Performance:** Maintained stable 60 FPS on mid-range Android devices (Snapdragon 665+) with 2,000+ on-screen animated entities.

Block Blaster Extreme - Grid-Based Puzzle Game

Unity, C#, Editor Scripting, Firebase

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- Custom Level Editor:** Developed a Unity Editor Window (LevelMakerTool) with paint-based grid interface and ScriptableObject serialization - reduced level creation time by 90% (from ~20 min to <2 min per level).

- **Rendering Optimization:** Used MaterialPropertyBlock for dynamic block/cannon color changes, avoiding runtime Material instantiation - prevented memory spikes and preserved draw-call batching on mobile.
- **Modular Architecture:** Engineered decoupled systems (GridManager, InventoryManager, SlotManager) using Observer pattern and event-driven communication - zero tight coupling between subsystems.
- **LiveOps:** Integrated Firebase Anonymous Auth for frictionless onboarding + Remote Config for server-side daily reward balancing without app updates.

Tri-Language Dictionary - Utility App

Unity, C#, Android

Solo Developer · 1,000+ Organic Downloads · [Google Play](#)

- **Memory-Efficient Audio:** Engineered Lazy Loading audio system that streams clips from disk on-demand - reduced runtime memory footprint by ~60% vs. pre-loading.
- **Data Architecture:** Built flexible CSV parser converting raw datasets into custom C# objects, enabling bidirectional search across 3 languages with offline caching.
- **Full Product Lifecycle:** Managed concept , design , development , release and iteration. Iterated on UI/UX based on user reviews, achieving 1,000+ organic downloads and 4+ star rating.

PROFESSIONAL EXPERIENCE

Unity Game Developer (Indie)

2021 – Present

Squaresum Games · Giza, Egypt

- Managed the full product lifecycle for multiple mobile titles - from architecture design through Google Play release, ASO optimization, and post-launch crash analytics.
- Achieved 1,000+ organic users on a published title through UI/UX iteration driven by Play Console metrics and user feedback.
- Integrated monetization pipelines (AdMob/Unity Ads) and tracked retention KPIs using Google Play Console and Firebase Analytics.
- Optimized all titles to maintain 60 FPS on low-end Android devices (2GB RAM, Adreno 505) while preserving visual fidelity through custom shaders and batching strategies.
- Implemented CI-adjacent workflows: version control with Git, structured branching, and build automation for Android APK/AAB generation.

EDUCATION

B.Sc. Communication & Electronics Engineering

2014 – 2019

Nile University · Giza, Egypt

GPA: 3.52/4.0 · Dean's Honor List · Coursework: Linear Algebra, Signal Processing, Embedded Systems, Data Structures & Algorithms